



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SREE ARASU AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 134.23 km
Estimated Duration: 3.0 hours
Adjusted Duration (Heavy Vehicle): 3.8 hours
Start: (11.4381, 77.8734)
End: (12.053992, 78.683811)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map: The route from CVQF+23W, Sangagiri to 3M3M+HH3, Naripalli spans approximately 134.23 kilometers, primarily traversing the state highways and rural roads of Tamil Nadu. The journey typically involves moving west from Sangagiri into rural areas, through Erode and closer to

Salem, before approaching Naripalli. This route includes travelling on NH44, which is an important north-south corridor, contributing to a mix of urban, semi-urban, and rural driving environments.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu experiences a tropical climate, with hot and humid summers and moderate monsoon rains. During the monsoon season (June to September), heavy rains can lead to poor visibility and slippery roads. Seasonal fog during the early winter mornings can also affect visibility. It is crucial to monitor weather updates via local forecasting services to avoid any weather-related disruptions.

3. Analysis of Traffic Patterns: Traffic congestion is typically higher during early morning (7–9 AM) and late evening (5–7 PM) rush hours, particularly around urban centers such as Erode and Salem. Industrial zones near these towns might also contribute to higher traffic, including heavy commercial vehicles. The NH44 and major urban intersections can be congested during these times.

4. Assessment of Road Quality and Infrastructure: The NH44 and state highways are generally well-maintained, offering multiple lanes and proper signage. However, off these major roads, drivers might encounter narrower, single-lane roads with mixed conditions in terms of maintenance. Potholes and temporary repairs in rural segments require vigilance.

5. Suggestions for Alternative Routes for Emergencies: In cases of blockages or emergencies, detours closer to Erode and Salem via local roads are possible but may add to travel time. Utilizing NH544 near Erode or NH179A near Salem can serve as emergency detour options. It's essential for drivers to have local maps or GPS guidance to navigate these detours.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Transport of hazardous materials in Tamil Nadu requires adherence to both state and national regulations. This includes having proper vehicle documentation, placarding, driver qualifications, and permissions from regulatory bodies. Compliance with local timings and routes designated for hazardous materials is mandatory.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials: Historically, incidents involving heavy vehicles in Tamil Nadu have been reported due to human error, poor visibility, and adverse weather conditions. Proper risk management and driving during favorable conditions can mitigate such risks. No major specific incidents on this particular route were recorded, but standard caution advised.

8. Environmental Considerations and Sensitive Areas: The route passes through agricultural regions which require careful navigation, especially during peak farming seasons. Attention to avoiding spillages and adherence to environmental regulations are crucial. Water bodies in the region should be protected from contamination.

9. Analysis of Communication Coverage: Major roads like NH44 generally have good cellular coverage. However, rural roads may experience patchy signals, especially in less populated or hilly areas. Having a reliable communication device or satellite phone is recommended for emergency communications.

10. Estimated Emergency Response Times: Emergency response times may vary from 30 to 60 minutes in urban areas near Erode and Salem and might extend up to 1-2 hours in remote rural areas due to limited emergency service coverage. Pre-identified emergency contact points and routes should be part of the travel planning.

12. Overall Summary of Risk Assessment: While the route poses moderate risks with respect to traffic congestion and weather conditions, it maintains a generally favorable safety profile due to well-developed infrastructure in many segments. Compliance with regulations, awareness of environmental considerations, and preparedness for communication and emergency scenarios are necessary to minimize risks. Overall, the route is manageable for trained drivers with proper precautionary measures.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45028, 77.87202	15 KM/Hr
5	Turn	Medium	11.46400, 77.87323	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	High	11.49149, 77.89601	15 KM/Hr
8	Turn	High	11.49224, 77.89601	15 KM/Hr
9	Turn	Medium	11.53501, 77.95707	30 KM/Hr
10	Turn	Medium	11.53601, 77.95856	30 KM/Hr
11	Turn	Medium	11.62403, 78.10893	30 KM/Hr
12	Turn	High	11.64172, 78.11817	15 KM/Hr
13	Turn	Medium	11.64224, 78.11928	30 KM/Hr
14	Turn	High	11.64195, 78.12129	15 KM/Hr
15	Turn	High	11.64120, 78.12168	15 KM/Hr
16	Turn	Medium	11.63948, 78.12081	30 KM/Hr
17	Turn	Medium	11.62861, 78.16526	30 KM/Hr
18	Turn	Medium	11.65703, 78.20465	30 KM/Hr
19	Turn	Medium	11.66124, 78.21011	30 KM/Hr
20	Turn	Medium	11.66732, 78.25597	30 KM/Hr
21	Turn	Medium	11.66639, 78.32382	30 KM/Hr
22	Turn	High	11.65716, 78.40570	15 KM/Hr
23	Turn	High	11.70480, 78.41538	15 KM/Hr
24	Turn	High	11.70565, 78.41336	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
25	Turn	Medium	11.70603, 78.41342	30 KM/Hr
26	Turn	High	11.70683, 78.41406	15 KM/Hr
27	Turn	Medium	11.71553, 78.44012	30 KM/Hr
28	Turn	Medium	11.72306, 78.44868	30 KM/Hr
29	Turn	Medium	11.72326, 78.44991	30 KM/Hr
30	Turn	Medium	11.72519, 78.45229	30 KM/Hr
31	Turn	Medium	11.74384, 78.46476	30 KM/Hr
32	Turn	High	11.74536, 78.48955	15 KM/Hr
33	Turn	Medium	11.75251, 78.49129	30 KM/Hr
34	Turn	Medium	11.77193, 78.50592	30 KM/Hr
35	Turn	High	11.77982, 78.51613	15 KM/Hr
36	Turn	Medium	11.77953, 78.51819	30 KM/Hr
37	Turn	Medium	11.77981, 78.51949	30 KM/Hr
38	Turn	Medium	11.78396, 78.52260	30 KM/Hr
39	Turn	Medium	11.79658, 78.52952	30 KM/Hr
40	Turn	Medium	11.79904, 78.53047	30 KM/Hr
41	Turn	Medium	11.82082, 78.55145	30 KM/Hr
42	Turn	Medium	11.82794, 78.55448	30 KM/Hr
43	Turn	Medium	11.87295, 78.60077	30 KM/Hr
44	Turn	Medium	11.87312, 78.60164	30 KM/Hr
45	Turn	High	11.87362, 78.60226	15 KM/Hr
46	Turn	Medium	11.87456, 78.60228	30 KM/Hr
47	Turn	Medium	11.88169, 78.61312	30 KM/Hr
48	Turn	Medium	11.93519, 78.64218	30 KM/Hr
49	Turn	Medium	11.93806, 78.64234	30 KM/Hr
50	Turn	Medium	11.97407, 78.66551	30 KM/Hr
51	Turn	High	11.97462, 78.66619	15 KM/Hr
52	Turn	Medium	11.97450, 78.66716	30 KM/Hr
53	Turn	Medium	11.97526, 78.66842	30 KM/Hr
54	Turn	High	11.98467, 78.67005	15 KM/Hr
55	Turn	Medium	11.98539, 78.66855	30 KM/Hr
56	Turn	Medium	11.99282, 78.66631	30 KM/Hr
57	Turn	High	12.01658, 78.66272	15 KM/Hr
58	Turn	High	12.01948, 78.67071	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
59	Turn	Medium	12.04599, 78.67217	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
14	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
21	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
22	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
23	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
24	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
28	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
30	hospital	Niharika Hospital	11.6565971, 78.3937356	30 km/h	Medium
31	hospital	Malar Hospital	11.6567854, 78.3928345	30 km/h	Medium
32	hospital	Sri Bala Hospital	11.6566667, 78.4010889	30 km/h	Medium
33	clinic	Sibi Physiotherapy Clinic	11.6561305, 78.3994263	30 km/h	Medium
34	hospital	Saraswathi Hospital, Valapady	11.6563505, 78.4009062	30 km/h	Medium
35	hospital	Sri Udaya Hospital	11.6560244, 78.4040862	30 km/h	Medium
36	hospital	Government Hospital, Valapady	11.65706, 78.403773	30 km/h	Medium
37	hospital	Devaki Hospital	11.6564189, 78.402376	30 km/h	Medium
38	hospital	Deepam Hospital	11.6576703, 78.4065155	30 km/h	Medium
39	hospital	Government hospital,Belur	11.70532, 78.419538	30 km/h	Medium
40	hospital	Tribal Hospital	11.9105443, 78.6261336	30 km/h	Medium
41	hospital	Government Hospital, Naripalli	12.0425018, 78.6693922	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
15	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
20	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
25	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
26	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
27	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
29	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62403, 78.10893



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64172, 78.11817



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64224, 78.11928



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64195, 78.12129



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64120, 78.12168



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63948, 78.12081



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.62861, 78.16526



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65703, 78.20465



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66732, 78.25597



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66639, 78.32382



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65716, 78.40570



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70565, 78.41336



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70603, 78.41342



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70683, 78.41406



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71553, 78.44012



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72306, 78.44868



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72326, 78.44991



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72519, 78.45229



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74384, 78.46476



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.74536, 78.48955



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.75251, 78.49129



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77193, 78.50592



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77982, 78.51613



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77953, 78.51819



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77981, 78.51949



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78396, 78.52260



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.79658, 78.52952



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.79904, 78.53047



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.82082, 78.55145



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.82794, 78.55448



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87295, 78.60077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87312, 78.60164



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.87362, 78.60226



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87456, 78.60228



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88169, 78.61312



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93519, 78.64218



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93806, 78.64234



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.97407, 78.66551



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.97462, 78.66619



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.97450, 78.66716



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.97526, 78.66842



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.98467, 78.67005



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.98539, 78.66855



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99282, 78.66631



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.01658, 78.66272



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04599, 78.67217

